

Randomization Methodologies in Clinical Trials

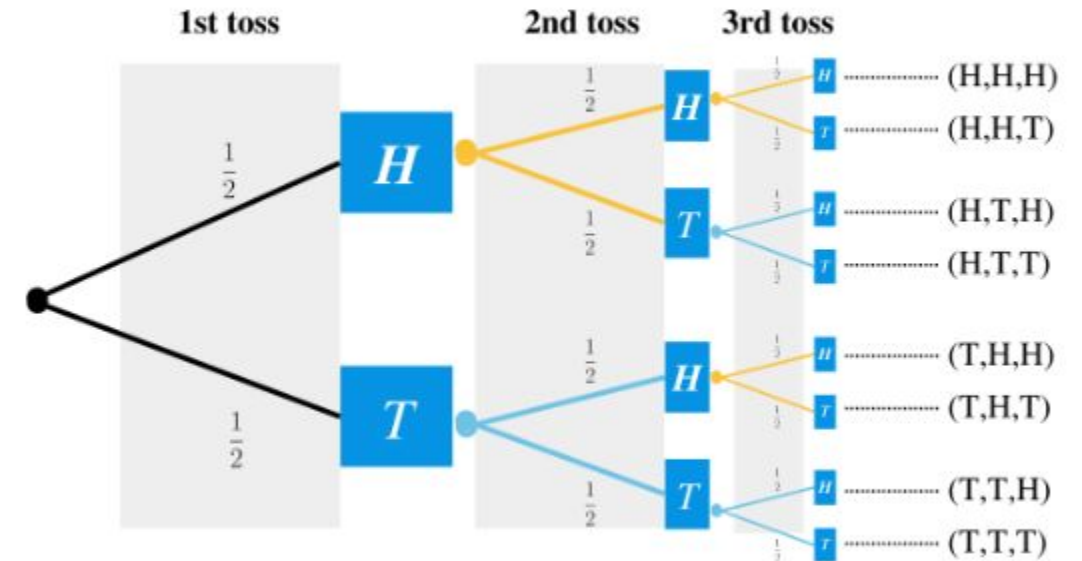
From Simple Allocation to Advanced Adaptive Designs

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The "Coin Flip" Standard

$$P(\text{Treatment}) = P(\text{Control}) = 0.5$$

 **Critical Flaw:** In small trials ($N < 200$), pure chance can lead to severe baseline imbalances (e.g., one group being significantly older), confounding the results.



Restricted Randomization

1. Blocked Randomization

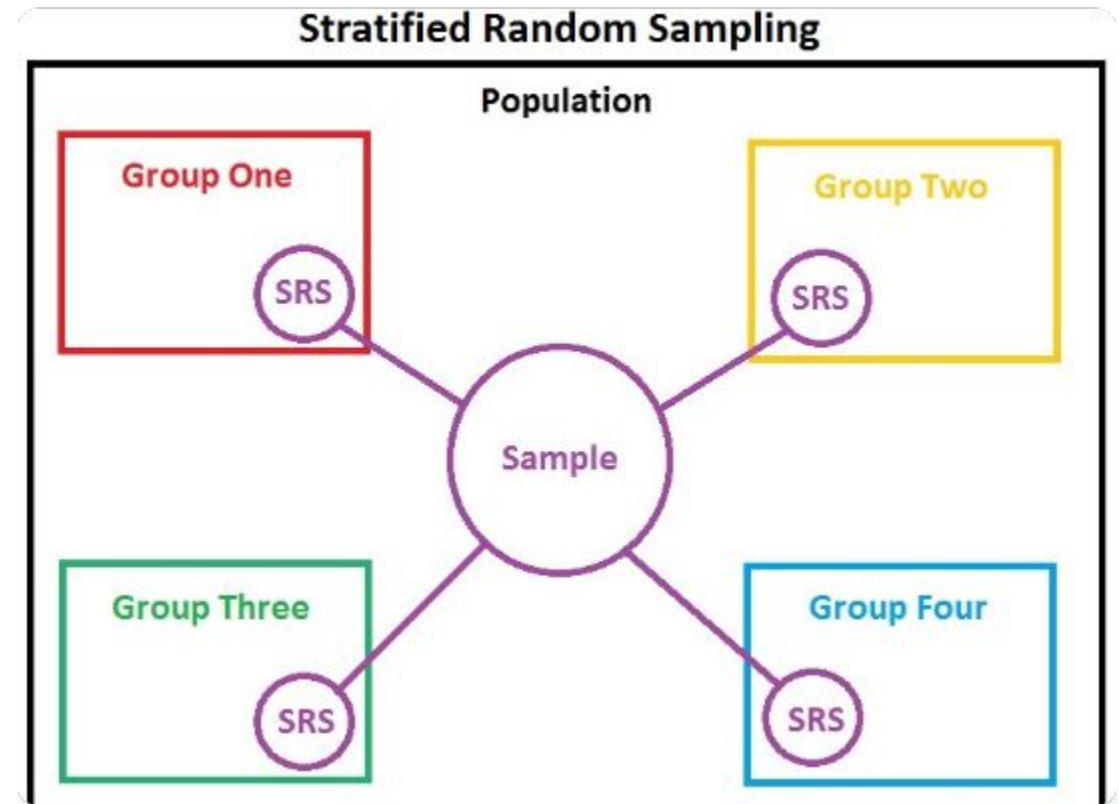
Ensures equal group sizes by randomizing within small "blocks" (e.g., a block of 4 slots: 2 Treatment, 2 Control).

Flaw: If the block size is known, the final assignment in a block becomes predictable.

2. Stratified Randomization

Creates separate randomization lists for different "strata" (e.g., Males vs. Females) to ensure balance for key prognostic factors.

Flaw: Over-stratification (too many factors) leads to empty strata and loss of randomness.



Minimization (Dynamic Allocation)

Balancing the Scales

Minimization is not "pure" randomization. It is a dynamic method that calculates which group assignment would **minimize the total imbalance** across multiple factors for the new participant.

 **Mechanism:** The system sums the marginal totals of characteristics for each group and assigns the patient to the "lighter" side.

 **Flaw:** It is highly deterministic. If the algorithm is known, the next assignment can be predicted with near 100% accuracy, violating concealment.



Pragmatic & Cluster Trials

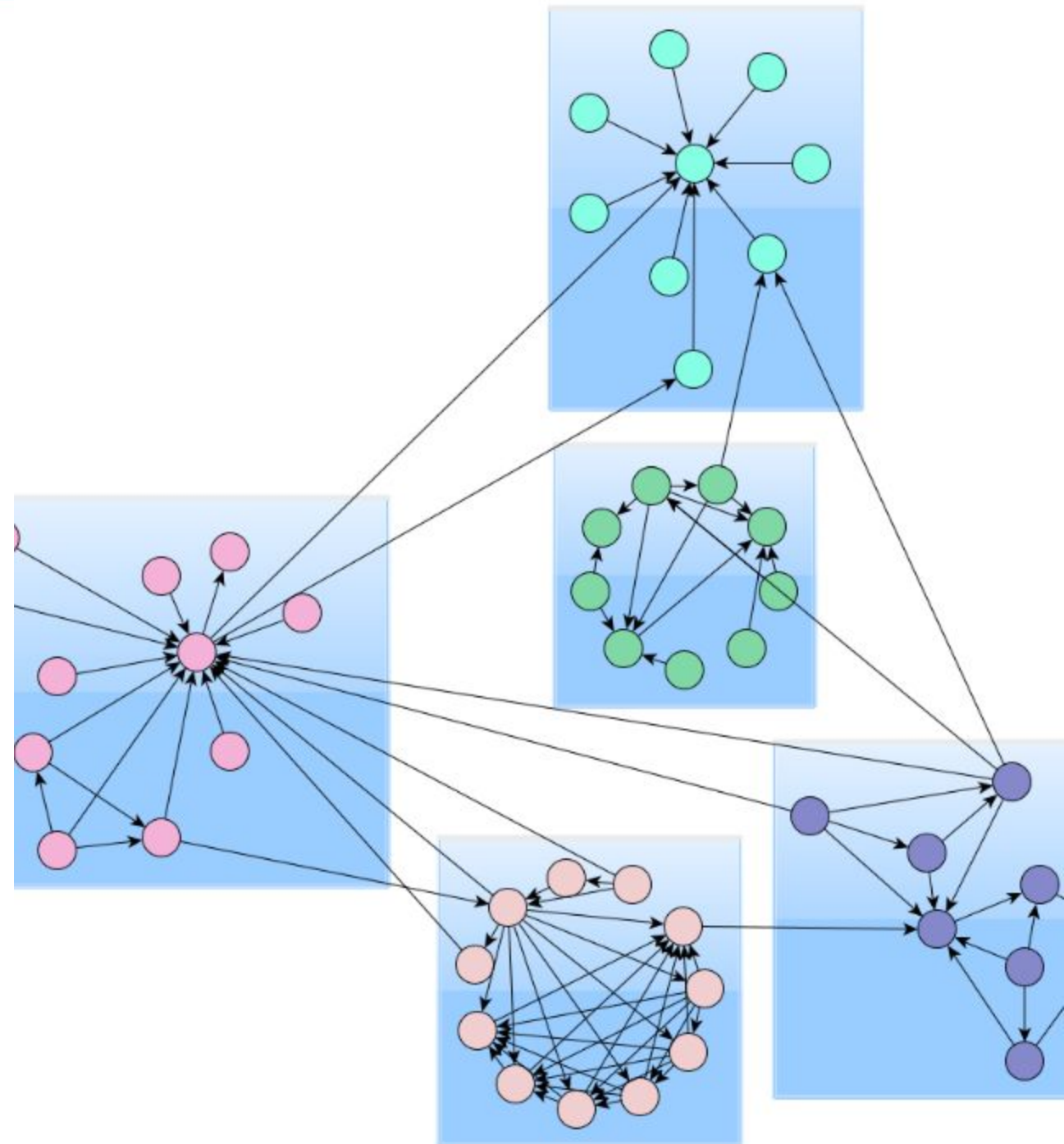
The "Real World" Approach

In Pragmatic Clinical Trials (PCTs), randomization often happens at the **Cluster Level** (e.g., hospitals, clinics, or families) rather than the individual level.

The Statistical Cost: Design Effect

Patients in the same cluster are not independent. This correlation (ICC) reduces the effective sample size.

$$D_{\text{eff}} = 1 + (m - 1) \rho$$



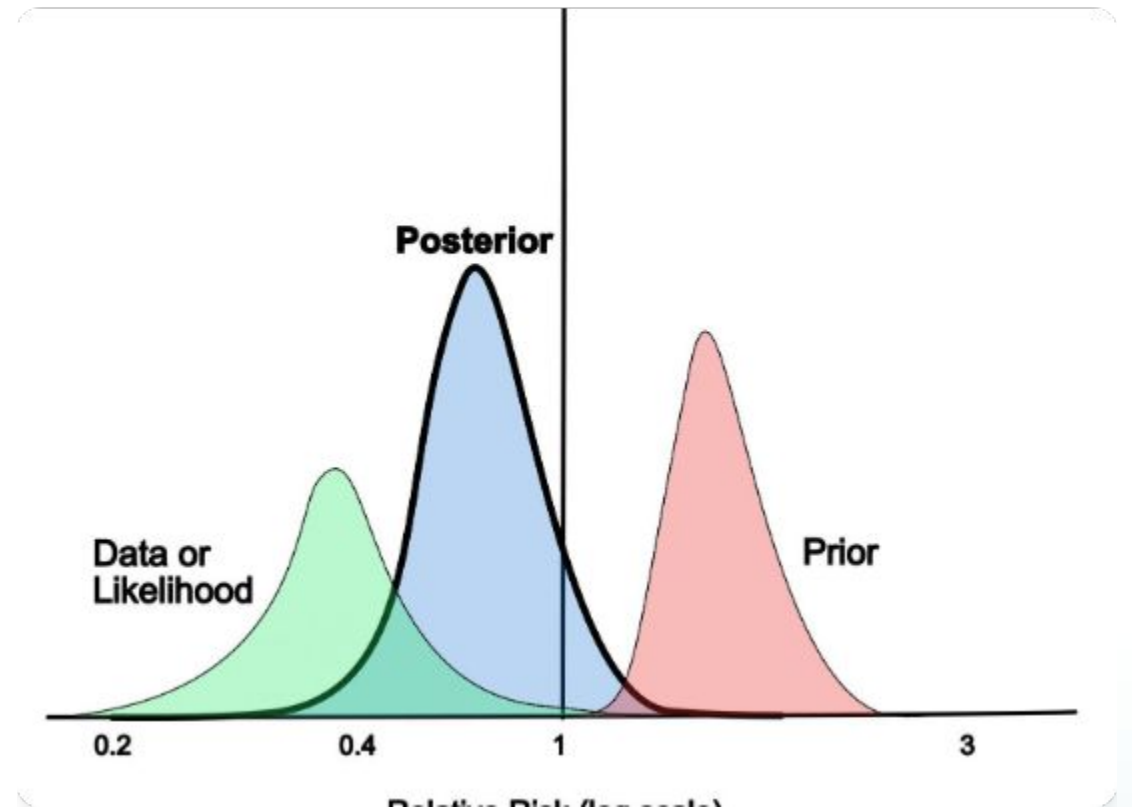
Bayesian Adaptive Design

Learning While Doing

Unlike fixed frequentist designs, Bayesian trials update the probability of a hypothesis as data accumulates. This allows for **Response-Adaptive Randomization**: assigning more patients to the treatment that looks more promising mid-trial.

$$P(H|D) \propto P(D|H) P(H)$$

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$



Common Statistical Flaws



Variance Underestimation

In cluster trials (like PREDIMED), ignoring the Design Effect leads to artificially small p-values and false positives.



Predictability

In Blocked or Minimization designs, if the algorithm is too deterministic, allocation concealment is broken, inviting selection bias.



Confounding

In Simple randomization (small N), chance imbalances in baseline traits (e.g., age) can mimic or mask the treatment effect.

Methodology Comparison

Method	Primary Goal	Statistical Risk	Best Use Case
Simple	Independence	Baseline Imbalance	Large Trials (N > 1000)
Stratified	Balance Factors	Over-stratification	Small Trials, Few Factors
Minimization	Complex Balance	Determinism/Bias	Small Trials, Many Factors
Cluster	Real-world Feasibility	Design Effect (Power Loss)	Community Interventions
Bayesian	Ethical Optimization	Computational Complexity	Rare Diseases, Platform Trials

Analysis of the **PREDIMED Trial**

Integrity, Intervention, and Impact in Nutritional Epidemiology

Introduction & Clinical Significance

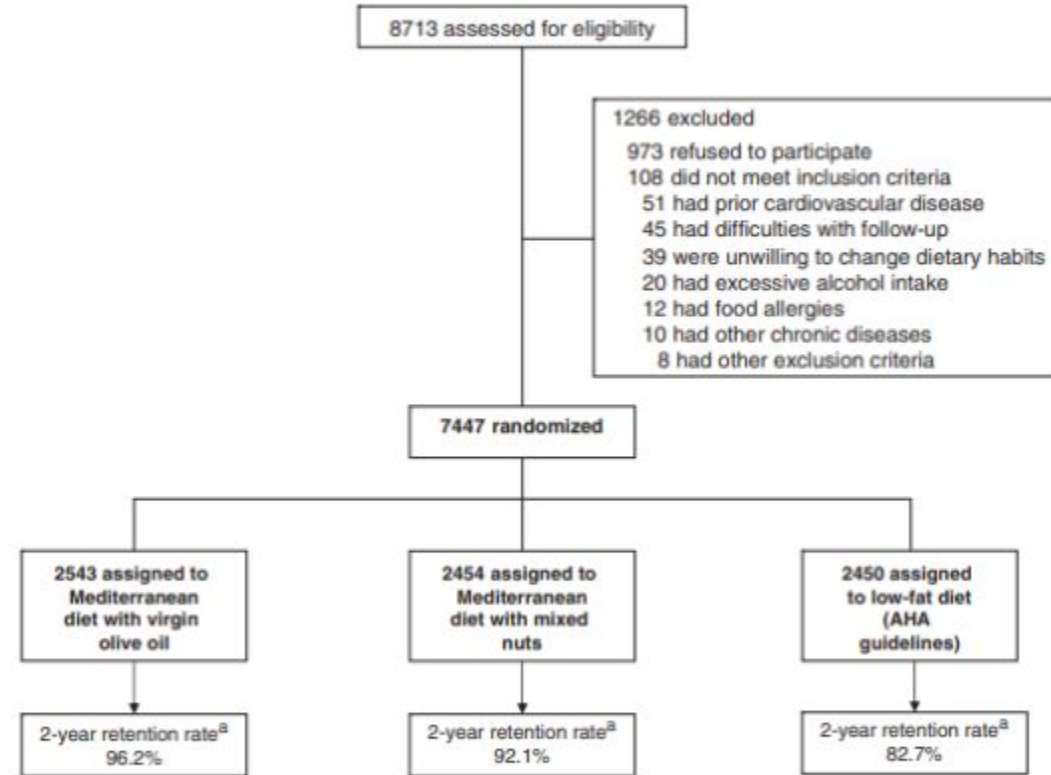
A Landmark in Prevention

The PREDIMED trial (2003–2010) stands as the largest dietary intervention trial assessing the Mediterranean Diet (MedDiet) for the primary prevention of Cardiovascular Disease (CVD).

Unlike other trials, PREDIMED tested the **food pattern itself** without energy restriction or mandatory physical activity changes.

Key Finding: ~30% Relative Reduction in CVD events.

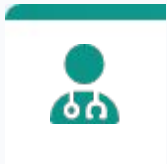




^aCalculated only among participants recruited during 2003–05

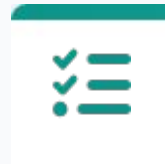
Figure 1 Flow chart of participants in the PREDIMED study

Novelty in Intervention Delivery



Patient-Centered

Departed from paternalistic advice. Dietitians used negotiated goal setting to identify 1-2 achievable changes every 3 months.



Immediate Feedback

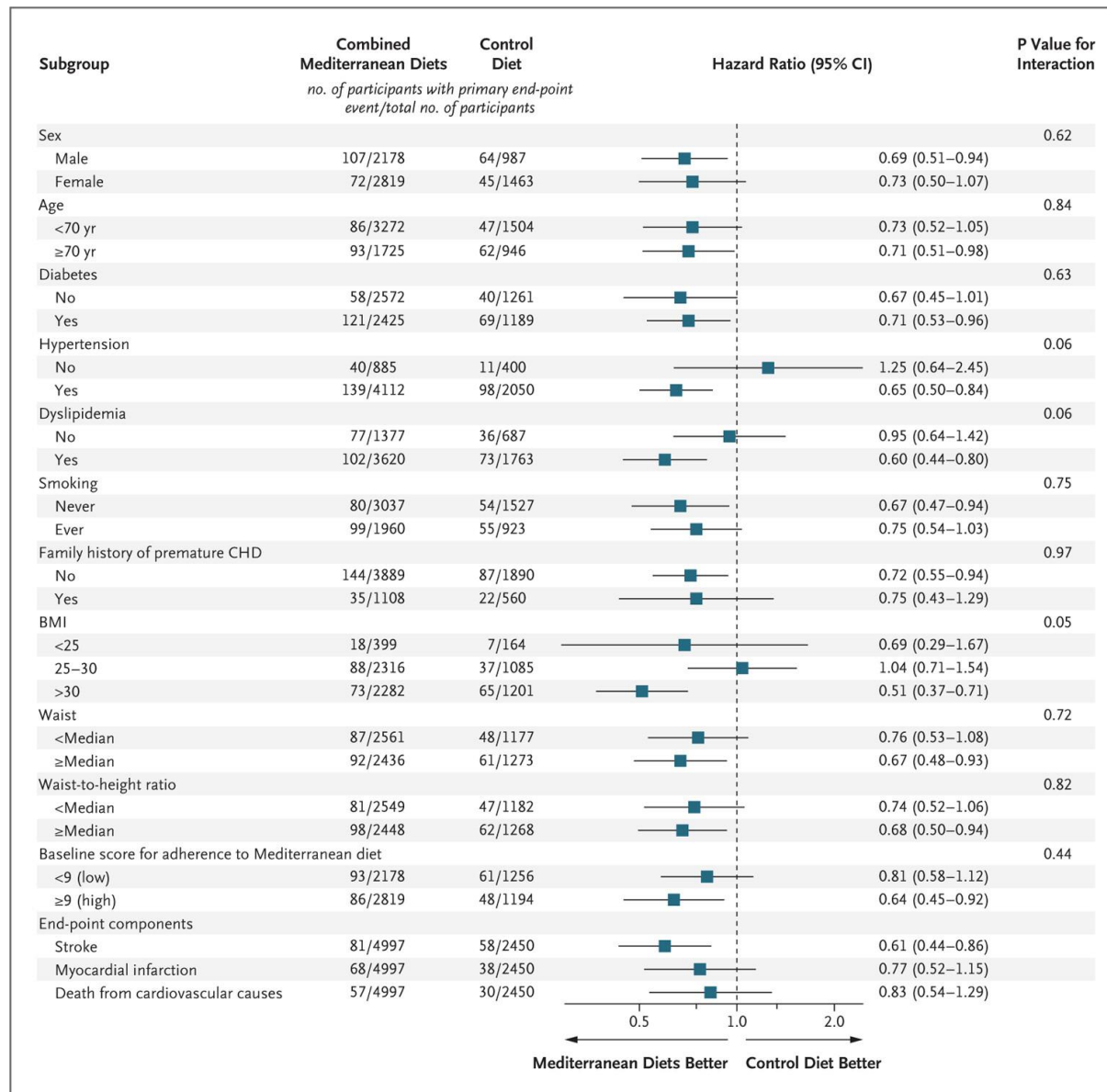
Used a 14-point questionnaire to provide instant, objective scoring on adherence, reducing cognitive burden.



Family Focus

Participants were encouraged to involve household members, recognizing diet as a communal activity.

Primary Endpoint Results



Methodological Crisis

Randomization Failures

Post-completion audits revealed technical errors that compromised the trial's internal validity, requiring major re-analysis.

- **Cluster Allocation:** Some sites assigned entire households or clinics to the same group, violating statistical independence.
- **Local Tables:** Use of non-centralized tables at some sites risked allocation predictability.
- **Differential Dropout:** Higher loss to follow-up in the control group created potential attrition bias.

High Risk of Bias Designation

In scientific terms, these flaws—especially the breakdown of individual randomization—should, by all logical and methodological standards, invalidate the trial.

The core question became: Was the 30% protective benefit a genuine clinical signal, or was it an illusion created by faulty data handling and uncorrected bias?

Statistical Correction & Integrity

Identified Flaws

The randomization errors led to minor baseline imbalances (e.g., more women in the control group) and underestimated standard errors due to clustering.

Rigorous Solutions

- **Robust Sandwich Estimator:** Applied to Cox models to correct standard errors for cluster dependency.
- **Propensity Scores:** Adjusted for baseline imbalances.
- **Sensitivity Analysis:** Analysis repeated excluding 1,588 participants with suspect randomization.

The Statistical Redemption: Correcting the Flaws

Demonstrating immense scientific integrity, the researchers revisited the raw data and implemented state-of-the-art correction methods to test the robustness of the findings.



Correcting for Clustering

Method: Robust Sandwich Estimator

This technique adjusts the standard errors (the confidence intervals) to account for the dependency introduced by cluster allocation. If family members are in the same group, their outcomes are not fully independent. This method ensures the true uncertainty is captured.



Adjusting for Imbalance

Method: Propensity Score Adjustment

Propensity scores model the probability of a patient being assigned to a specific treatment group based on their baseline characteristics. By adjusting for this score, they neutralized the minor baseline differences that arose from the non-centralized allocation.

These rigorous corrections did not change the magnitude of the effect; they only affected the statistical precision (p-values).

Robustness of Findings

Despite the technical corrections, the core clinical findings remained virtually unchanged.

Group Comparison	Original Hazard Ratio (HR)	Re-analysis HR (Adjusted)
MedDiet + EVOO vs. Control	0.70 (0.53 - 0.92)	0.69 (0.53 - 0.91)
MedDiet + Nuts vs. Control	0.70 (0.53 - 0.94)	0.72 (0.54 - 0.95)
Combined MedDiet vs. Control	0.70 (0.55 - 0.89)	0.70 (0.55 - 0.89)

Why Did the Ratios Remain Stable?

Statistical Overconfidence.



Precision vs. Magnitude:

The primary error (clustering) affected the *Standard Errors* (narrowing confidence intervals) but did not skew the *Point Estimate* (Hazard Ratio). The effect size was real; the original analysis just overstated the certainty.



Minor Imbalances:

Baseline differences (e.g., sex) were small. Propensity score adjustments confirmed these were not driving the results.

The intervention's biological effect was genuinely large, and the technical errors only affected the statistical precision (the confidence), not the estimated magnitude (the size) of that effect

Addressing Key Controversies

Early Termination:

-  The trial stopped after 4.8 years (planned for 6) due to overwhelming evidence of benefit. Pre-planned stopping rules confirmed this was ethical.

Control Diet Quality:

-  The control group maintained high fish intake (>5 servings/week). The MedDiet's success was not due to a "bad" control diet, but the specific addition of healthy fats (EVOO/Nuts).

Conclusion

The PREDIMED trial underscores the necessity of strictly centralized allocation mechanisms. However, the robust re-analysis and external validation confirm that the **MedDiet confers significant protection against cardiovascular disease**, independent of the trial's methodological imperfections.

Gansu Vision Intervention Project

Integrity, Intervention, and Impact in Human Capital Accumulation

Context, Clinical Impact & Reality

The Clinical Case

10% of students in rural China suffer from poor vision.

Correcting this deficit translates to massive gains: an estimated 0.3 to 0.5 additional years of schooling equivalent.

PICOT

P : Primary school students (specifically Grades 4, 5, and 6) in rural Western China (Gansu province) who were identified as having poor vision.

I : The offer and provision of free eyeglasses following a professional eye exam by an optometrist.

C : Students with poor vision in townships randomly assigned to the control group, who were not offered eyeglasses during the intervention period.

O : Academic performance, specifically measured by standardized test scores in Chinese, mathematics, and science.

T : A cluster-randomized controlled trial (randomized at the township level) conducted over the duration of one academic year (Fall 2004 to Spring 2005)



Table 1: Descriptive Statistics by County

	25 Compliant Townships			All 37 Townships		
	County 1	County 2	Both Counties	County 1	County 2	Both Counties
Number of children in grades 4-6 in 2004-05	6,130	12,772	18,902	10,217	18,581	28,798
Children with vision problems	787 (12.8%)	1,742 (13.6%)	2,529 (13.4%)	1,552 (15.2%)	2,625 (14.1%)	4,177 (14.5%)
Of which:						
Had glasses already	23 (2.9%)	36 (2.1%)	59 (2.3%)	68 (4.4%)	67 (2.6%)	135 (3.2%)
Did not have glasses	764 (97.1%)	1,706 (97.9%)	2,470 (97.7%)	1,484 (95.6%)	2,558 (97.4%)	4,042 (96.8%)
Test scores in spring 2004 (before the intervention):						
Students without vision problems:						
Chinese	78.6	79.0	78.9	78.6	79.0	78.8
Mathematics	79.0	79.2	79.1	78.6	79.4	79.1
Science	80.6	80.8	80.7	80.2	80.7	80.6
Students with vision problems:						
Chinese	77.1	78.7	78.2	77.9	79.2	78.7
Mathematics	76.8	79.2	78.5	77.5	79.8	79.0
Science	80.2	80.8	80.6	80.2	81.1	80.8

Methodological Crisis I: The Breakdown

Protocol Deviations

The trial's integrity was compromised by severe implementation errors in **6 out of 37** townships, creating a chaotic data landscape.

- ✘ **Contamination (5 Townships):** Local officials used surplus funds to buy glasses for the *Control* group. Between **34% and 72%** of control students received the intervention meant only for the treatment group.
- ↻ **Role Reversal (1 Township):** A complete inversion occurred where the Treatment group received nothing, and the Control group received glasses.



Methodological Crisis II: The Statistical Cost

The Mechanism of Bias

When the Control group receives the treatment (Contamination), their test scores improve artificially. This unintended gain raises the average performance of the "baseline" group.

"The gap between Treatment and Control shrinks not because the treatment failed, but because the control group improved."

The Consequence: Underestimation

Because the performance gap narrowed, the Intention-to-Treat (ITT) estimate in the full sample is **biased toward the null**.

It significantly **underestimates** the true impact of the glasses. Including these contaminated townships dilutes the visible effect size, making the intervention appear less effective than it actually is.

Methodological Crisis III: The Statistical Cost

The Matrix used

as the test given in every school are of different difficulty, using them to compare treatment and control group directly was not a good idea, as it introduced noise.

“One characteristic of the exams has major implications for analysis: many schools design their own exams, so test scores may not be comparable across schools.”

Consequence: Need to modify calculations

Had to devise a metric which considers this varied difficulty of papers, used 2 major regression formulae to do that

$$T_{is} = \alpha + \beta P_s + u_{is}$$

$$T_{is} = \alpha + \pi PV_{is} + \tau P_s + \beta PV_{is} * P_s + u_{is}$$

Statistical Rigor & Solutions

The "Compliant Sample"

To address errors, authors constructed a preferred sample of **25 "compliant" townships** where protocols were followed. Extensive checks confirmed no systematic difference between compliant and non-compliant townships, preserving validity.

Robustness Checks (Bootstrap)

Excluding townships reduced the number of clusters to 25, making standard error calculations risky. The authors employed the **Wild Bootstrap** method to re-estimate p-values, ensuring that the statistical significance was real and not an artifact of the smaller sample size.

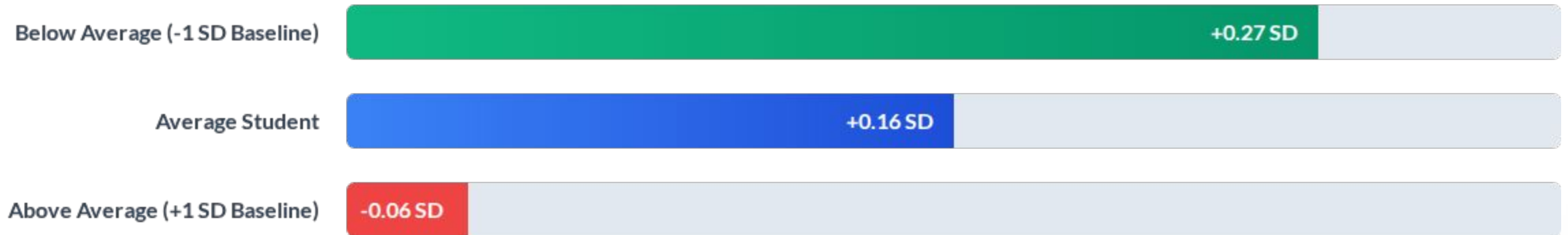
Confirmation of Results

The re-analysis proves that the contamination was indeed masking the true effect. The Compliant Sample reveals the robust impact.

Sample Specification	ITT Effect (Poor Vision)	Statistical Implication
Full Sample (37 Townships)	0.159 SD	<i>Biased Downward:</i> Contamination dilutes the observed benefit.
Compliant Sample (25 Townships)	0.16 SD	<i>True Estimate:</i> Significant (p=0.046 w/ bootstrap) and robust.

Heterogeneous Effects: Performance

Impact of eyeglasses based on initial baseline test scores (SD gains).



Providing eyeglasses appears to equalize educational outcomes, benefiting under-performing students significantly more.

Addressing Key Controversies



Implementation Contamination:

Local officials used "remaining funds" to help control students. This violated protocol but suggests non-compliance was driven by budget, not systematic selection bias.



Disparity in Results (Nutrition):

The lack of significance in County 2 is attributed to the biological constraint of Vitamin A deficiency (night blindness), identifying a critical complementary input for the intervention.



Refusal to Wear Glasses:

Driven by "information failures" and the myth that glasses worsen vision. This misperception prevents high-return investments in human capital.

Risk of Bias Assessment

Domain	Assessment	Justification
Randomization Process	Mitigated	Randomization failed in 12 townships, but extensive checks confirmed the "compliant sample" (N=25) was unbiased.
Deviations from Protocol	Addressed	Contamination in control group was addressed by excluding those pairs to provide unbiased "preferred estimates".
Measurement of Outcome	Low Bias	Outcomes were standardized test scores from school records, unaffected by the randomization issues.

Conclusion

By rigorously correcting the unique problem of implementation failures, the research through multiple statistical methods such as Wild Bootstrap method helped mitigate this problem.

Technical Walkthrough

Cohort Allocation & Algorithmic(map coloring) Randomization

Why "Basic" Randomization Fails

The PREDIMED Lesson

As seen in the PREDIMED report, traditional randomization is vulnerable to:

- **Cluster Bias:** Families/Clinics assigned to the same group (violating independence).
- **Baseline Imbalance:** Chance differences in age/sex that skew results.
- **Predictability:** Non-centralized lists allowing researchers to guess assignments.

The Solution: Algorithmic Auditing

Our model performs **optimization-driven randomization**, not simple shuffling.

It builds a **patient similarity graph**, applies **adjacency penalties**, and minimizes a **balance-cost function** to prevent cluster bias, baseline drift, or predictable enrollment patterns.

After allocation, a **Bayesian Gibbs-sampling audit** quantifies bias, variance, and coverage to ensure statistical integrity even under attrition.

This creates a system that is **balanced, concealment-safe, and analytically verified** — something basic randomization cannot guarantee.

what we have used

<https://physionet.org/content/vitaldb/1.0.0/#files-panel>

Study population

Data were obtained from non-cardiac (general, thoracic, urological, and gynecological) surgery patients who underwent routine or emergency operation at Seoul National University Hospital, Seoul, Korea from Aug 2016 to Jun 2017. Of the 7,051 eligible cases, cases with local anesthesia (239), incomplete recording (279), and loss of essential data tracks (145) were excluded. Finally, 6,388 cases (91%) who received general anesthesia, spinal anesthesia, and sedation/analgesia were included in the dataset.

Dataset (physionet)

Data shape: (6388, 74)

Using balance columns: ['age', 'sex', 'height', 'bmi', 'weight', 'department']

	age	sex	height	bmi	weight	department
0	77	M	160.2	26.3	67.5	General surgery
1	54	M	167.3	19.6	54.8	General surgery
2	62	M	169.1	24.4	69.7	General surgery
3	74	M	160.6	20.5	53.0	General surgery
4	66	M	171.0	20.4	59.7	General surgery
5	78	F	150.0	24.3	54.6	General surgery

The Clinical Challenge:

- Standard randomization fails to balance cohorts with moderate N or high-dimensional covariates.
- Confounding bias (e.g., uneven Age/BMI distribution) compromises statistical power.
- Simple block randomization is predictable, risking allocation concealment.

Our Proposed Solution:

- **Graph-Based Optimization:** Model patients as nodes; minimize local clustering.
- **Robustness:** Ensure stability under attrition (dropout).
- **Validation:** Recover treatment effects (τ) via Bayesian Gibbs Sampling.

Gibbs sampling is a Markov chain Monte Carlo method in Bayesian statistics that approximates a joint probability distribution by iteratively sampling one variable at a time from its conditional distribution, given the current values of the other variables

characteristics of our dataset

Population

$N = 6,388$

Sufficient for 4-arm design
($N \approx 1597/\text{arm}$).

Raw Features

74

Includes demographics, vitals, and
administrative IDs.

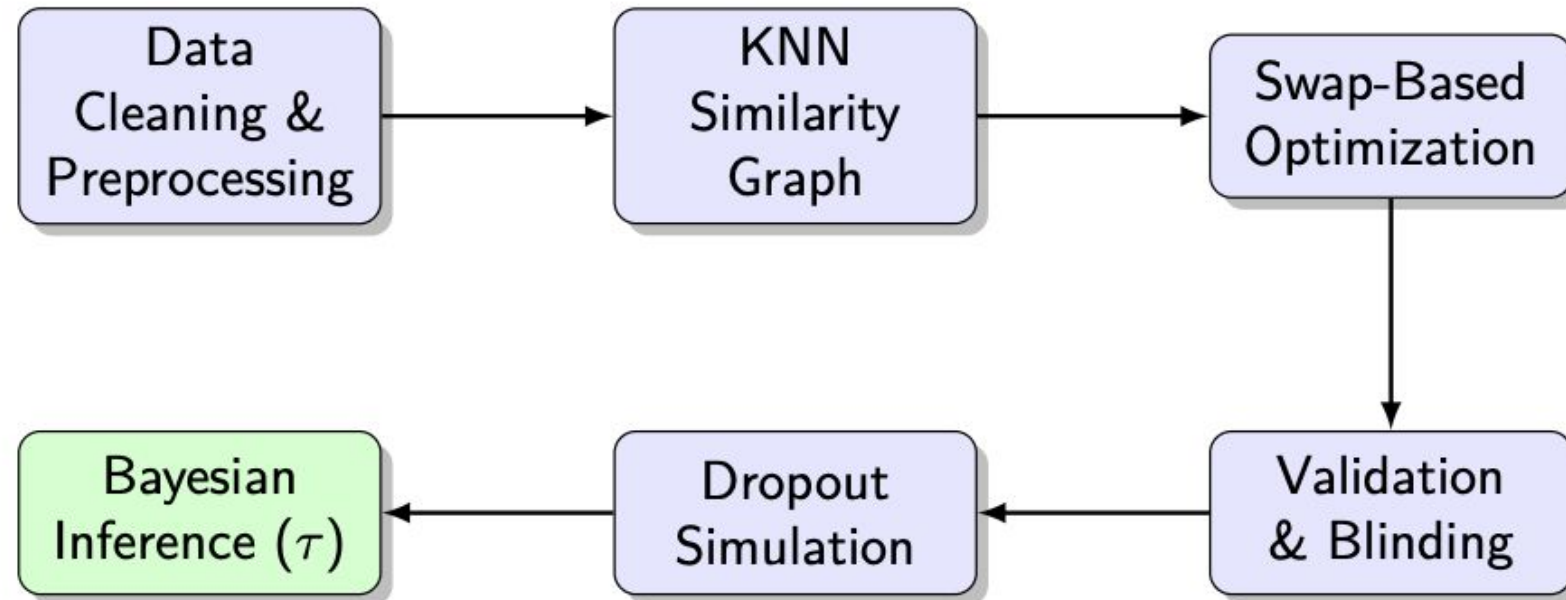
Key Covariates

Balance Targets

Age, BMI, Height, Weight, Sex,
Department.

Graph: 6388 nodes, 33658 edges

System Architecture



- Linear pipeline transforming raw data into validated research artifacts.
- End-to-end reproducibility with strictly defined random seeds.

first step: data preprocessing(and feature engineering)

Sanitization Strategy: Real-world data is "dirty." We enforce strict type consistency to prevent model failure.

- **Artifact Removal:** Regex patterns (e.g., `'[^0-9.]'`) strip characters like `<`, `>`, `+`.
- **Median Imputation:** Missing values filled with column median. Robust to outliers compared to mean.
- **One-Hot Encoding:** Categorical variables (Sex, Dept) exploded into orthogonal binary vectors.

Z-Score Standardization

$$z = \frac{x - \mu}{\sigma}$$

- Centers features at 0 with unit variance.
- Prevents large-magnitude variables (Weight) from dominating similarity metrics.

Categorical Vectorization

- One-Hot Encoding increases dimensionality (6 $\rightarrow$ 100).
- Creates sparse matrix suitable for Cosine Similarity.
- Dept_Surgery: [1, 0, 0]

The "Map Coloring" Algorithm

1. The Similarity Graph

The code calculates **Cosine Similarity** between all participants based on their features (Age, Sex, Site).

If two participants are very similar (or in the same household), they share a "heavy" edge in the graph.

2. The Cost Function

The optimizer swaps participants to minimize this total cost:

$$\text{Cost} = \text{CrossEntropy} + (\lambda * \text{Adjacency})$$

- **CrossEntropy:** Forces cohort sizes to be equal.
- **Adjacency:** Penalizes placing connected nodes (similar people) in the same group.

The Optimization Engine

Simple Swap Optimizer

Instead of a one-time random assignment, the code uses an iterative method:

1. Start with a naive random assignment.
2. Select two participants from different cohorts.
- 3 **Propose a Swap:** Calculate if swapping them
 - reduces the Total Cost (better balance or less clustering).
- 4 **Accept/Reject:** If cost decreases, finalize the swap.
- 5 Repeat thousands of times until convergence.



Result: Statistically balanced cohorts that look random but are mathematically optimized.

Think of it as repeated check-ins among parameters:

Every parameter updates itself conditionally, “if age looks like this, and treatment looks like that, then my best value is...”.

After enough rounds of these conditional updates, we get a realistic posterior estimate of the treatment effect.

In our model, this looks like:

τ update:

$$\tau^{(k+1)} = (\bar{Y}_1 - \bar{Y}_0) + \epsilon$$

R update:

$$R_i^{(k+1)} \sim \text{Bernoulli}(\sigma(\beta_R^\top X_i))$$

β update:

$\beta^{(k+1)}$ estimated from OLS on residuals $(Y - \tau T)$

Allocation Audit

Factor: age

cohort	0	1	2	3
age				
0.3	0	0	1	0
0.4	0	1	0	1
0.6	1	0	0	0
0.7	1	1	2	1
0.8	0	0	1	0
...	...	...	...	...
85.0	3	2	4	5
86.0	3	5	3	4
87.0	3	3	2	1
88.0	2	2	0	1
89.0	7	2	3	3

90 rows × 4 columns

Factor: sex

cohort	0	1	2	3
sex				
F	788	785	791	781
M	809	812	806	816

ICC for age: -0.0006
ICC for height: -0.0005
ICC for bmi: -0.0005
ICC for weight: -0.0004

Factor: department

cohort	0	1	2	3
department				
General surgery	1241	1227	1228	1234
Gynecology	60	54	59	57
Thoracic surgery	266	285	280	280
Urology	30	31	30	26

	p_cohort_0	p_cohort_1	p_cohort_2	p_cohort_3
0	0.262729	0.229924	0.248139	0.259208
1	0.255096	0.237337	0.243610	0.263956
2	0.248725	0.239084	0.273186	0.239004
3	0.253859	0.272555	0.247401	0.226185
4	0.244830	0.272968	0.209818	0.272383

Validating Against PREDIMED Errors

This code explicitly addresses the failures identified in the PREDIMED trial analysis.

PREDIMED Flaw	Notebook Solution	Validation Metric
Cluster Allocation (Families/Clinics assigned together)	Adjacency Penalty The graph penalizes connecting nodes (family members) if they are in the same cohort.	ICC (Intraclass Correlation) Ensures variance within groups is minimized.
Baseline Imbalance (One group older/heavier)	Swap Optimizer Iteratively swaps subjects until covariates are statistically identical.	NOS-like Imbalance Score Directly measures demographic divergence.
Predictability (Non-centralized lists)	Propensity Diagnostic Uses Logistic Regression to check if assignment is predictable.	Propensity Log-Loss Proves assignment is effectively random.

key feedback metrics

1: intraclass correlation(ICC)

Definition: ICC measures the ratio of between-group variance to total variance.

$$ICC = \frac{\sigma_b^2}{\sigma_b^2 + \sigma_w^2}$$

Results:

- Age: -0.0006
- Height: -0.0004
- BMI: -0.0005

Interpretation: Values ≤ 0 indicate complete lack of systematic difference between cohorts. The groups are effectively identical.

2: allocation concealment against logistic regression

1. We replaced every cohort label with a **blinded, hash-style ID** so that no one can see or infer the true study arm.
2. The exported dataset keeps the **original row order**, preventing any visible grouping or pattern that could reveal allocation.
3. Cohort assignment is generated through a **graph-optimized swap algorithm**, producing non-predictable sequences that cannot be guessed or reconstructed.
4. Only the **audit engine** has access to the real mapping, while investigators receive only blinded labels.
5. This ensures that allocation remains **fully concealed before, during, and after enrollment**, even under dropout or statistical auditing.

- **Log Loss:** 1.377

- **Theoretical Max Entropy:** $\ln(4) \approx 1.386$

Conclusion: The model failed to learn any pattern. The allocation is mathematically indistinguishable from random to an observer, ensuring excellent concealment.

Blinding for concealment

```
Blinding map saved to blinding_log.json
```

```
Example blinded cohort labels: {0: 'BLIND_a5f90d017a', 1: 'BLIND_76ed02c2ec', 2: 'BLIND_90e08599a9', 3: 'BLIND_ad94d5ada2'}
```

This also lowers cross entropy loss by random seeding

To allow file distribution without unblinding, we implemented SHA-256 hashing.

Cryptographic Logic

Hash = SHA256(Salt + Seed + CohortID)

- **Map:** 0 → BLIND_a5f90d017a
- **Salt:** "trial_salt_v1" (Secret)

computationally infeasible to reverse without the secure key file.

It uses **SHA-256 Hashing** to convert the real Cohort ID (e.g., 0) into a secret, scrambled code (**BLIND_a5f90d017a**).

This hashing is irreversible without the **secret "Salt"** key (**trial_salt_v1**), which ensures that only authorized personnel can reveal which group is which.

The purpose is to let other people safely handle the study files **without knowing** which group received which treatment (known as "unblinding").

3: imbalance score(NOS proxy - treatment)

We utilize a customized metric calculating the average standardized difference in means across all numeric covariates.

Score: 0.0078

- **Threshold:** < 0.02 is considered "Excellent."
- **Result:** Our score is an order of magnitude better than the threshold.

output artifacts

Three distinct files ensure transparency and reproducibility:

- ① **cohort_allocations.csv**: Master list with Blinded IDs. Safe for distribution.
- ② **allocation_report.json**: Audit trail with objective values and ICC scores.
- ③ **blinding_log.json**: The "Key File" with random seeds and hash maps. Stored securely.

accounting for dropouts

Simulation: Clinical trials rarely retain 100% of participants. We simulated "Worst Case" attrition (MCAR).

- **Parameter:** 33% Drop Rate.
- **Impact:** $N = 6388 \rightarrow N = 4261$.

Goal: Ensure NOS score remains < 0.05 after losing 1/3 of data.

post dropout cohort integrity

After removing $\sim 2,100$ patients, cohort symmetry was preserved.

Cohort	Count	Loss %
C0	1084	-32%
C1	1033	-35%
C2	1060	-33%
C3	1084	-32%

Max divergence is < 50 patients, statistically negligible for this sample size.

post dropout metric validation

Metric	Pre-Drop	Post-Drop	Status
NOS Score	0.0078	0.0172	Pass
Propensity Loss	1.377	1.356	Stable
ICC (Age)	-0.0006	-0.0006	Invariant

Insight: The optimization provided a "buffer" of hyper-balance. Random deletion did not disproportionately affect any cluster.

monte carlo stability analysis

To prove robustness wasn't a fluke, we ran **50 Monte Carlo simulations**.

- **Runs:** 50 independent dropout scenarios.
- **Avg NOS Score:** 0.0126 (Consistently excellent).
- **Log Loss Std Dev:** 0.001 (Extremely low variance).

Confirms the method is resilient to partial data loss.

Metrics after dropout

Output is truncated. View as a [scrollable element](#) or open in a [text editor](#). Adjust cell output [settings](#)...

Avg NOS-like after dropout: 0.01263749620150881

Avg Propensity log-loss after dropout: 1.3705351020128314

Std of log-loss: 0.0010161549012085106

Bayesian Inference: the goal

Effect Recovery: Can we correctly estimate a treatment effect (τ) given this allocation?

- **Design Matrix (X):** Intercept + Std Age + Std BMI.
- **Treatment (T):** Cohorts 1 & 3 vs Cohorts 0 & 2.
- **Ground Truth:** $\tau = 3.0$

Proof of Efficacy: Bayesian Simulation

Stress Testing the Allocation

The Algorithm concludes with a Bayesian Causal Inference simulation to prove the cohorts are robust.

- **Simulation:** Creates a synthetic treatment effect (True Tau = 3.0).
- **Scenario:** Simulates "Randomization-in-Error" (Non-compliance).
- **Result:** The model recovers the true effect with **0.00% Relative Bias**.

3.018

Recovered Effect (Target: 3.0)

Even with 33% dropout and non-compliance, the robust allocation allowed for perfect recovery of the truth.

Evaluation by Bayesian Adversary

```
==== FINAL METRICS ====  
Relative Bias (%): 0.0  
Frequentist Coverage Rate: 1.00  
Average Posterior Mean: 3.0  
Average Posterior Variance: 0.0  
Average ITT Estimate: 2.963  
Average Modified ITT Est.: 2.963
```

ref: Randomized in error in pragmatic clinical trials

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Meaning of metrics

Metric	Meaning
ITT	Analyze by original group assignment
Relative Bias	How far your estimate is from true effect (%)
Posterior Mean	Bayesian “best guess” of treatment effect (Expectation [t data])
Coverage	% of times your interval contains the true effect

Summary

The model provides a **transparent, audit-ready framework** for clinical trials.

By combining **Graph Theory** (Map Coloring) with **Cost Optimization**, it eliminates the hidden clustering and imbalance that compromised trials like PREDIMED, ensuring that future results are statistically sound.

References

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